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A fused convolutional transformer for voltammetric electronic tongue analysis tasks

Jingjing Liu [a,*], Jiangyong Li [a], Yifei Xu [a], Sze Shin Low [b], Haohao Ning [a], Shixin Yu [a], Dongfu Xu [a], Qingjun Liu [c,*]

[a] School of Automation Engineering, Northeast Electric Power University, Jilin 132012, China
[b] Research Centre of Life Science and HealthCare, China Beacons Institute, University of Nottingham Ningbo China, 199 Taikang East Road, Ningbo 315100, China
[c] Biosensor National Special Laboratory, Key Laboratory for Biomedical Engineering of Education Ministry, Department of Biomedical Engineering, Zhejiang University, Hangzhou 310027, China

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ABSTRACT

Electronic tongue and artificial intelligence have been evaluated as new rapid analysis technologies in complex solution research. The use of nanomaterials to modify the screen-printed carbon electrode (SPCE) increased the specific surface area and electrocatalytic active sites of the electrode. The voltammetric electronic tongue utilizes multiple cross sensitive sensors to describe the overall characteristics of complex samples, and the complementary information between sensors provides rich and comprehensive descriptions for analyzing various components. The use of electronic tongue for rapid analysis of water quality has gradually become a trend, but the overlapping phenomenon of peaks formed by the similar redox potential of each component in water and the interference of the substrate makes it difficult to obtain effective information of each component in water quality. Here, we propose a Fused Convolutional Transformer model (FCvT) that removes the limitation of overlapping peaks by fusing local features and global complementary features to quantify components in complex solutions. The performance of FCvT was evaluated by testing synthetic water samples consisting of four standardized substance solutions and compared with partial least squares (PLS), support vector regression (SVR), artificial neural networks (ANN), and convolutional neural networks (CNN). The results show that the FCvT obtains the lowest mean absolute percentage error in the set of independent tests.The average quantization error of FCvT is reduced by 34.4 % relative to other methods.

1. Introduction

Electrochemical analytical methods provide convenient and rapid testing, and artificial intelligence plays an important role in the quantitative prediction of water quality parameters. Traditionally, analytical techniques for complex solutions are often complicated and cumbersome. Examples include traditional titration, spectrophotometry, ion chromatography, etc [1–3]. However, these methods require sample pretreatment, complex operating procedures, and bulky equipment. In contrast, electrochemical analysis provides a convenient and rapid means of detection [4]. The high sensitivity, selectivity, rapidity and ease of operation of electrochemical analysis techniques have been widely used in environmental monitoring. Making full use of electrochemical analysis techniques and sensor arrays in detection work has become the key to break through laboratory limitations and move towards field detection [5].

Electronic tongue systems usually consist of a series of low-selectivity sensors and data processing methods [6,7]. Sensor arrays with cross-sensitivity provide comprehensive responses to complex information about the sample. Multivariate data processing methods and pattern recognition methods establish the relationship between samples and data to complete tasks such as calibration and prediction [8–10]. The screen-printed carbon electrode (SPCE) modified with nanomaterials with extensive electrocatalytic activity has increased the specific surface area and electrocatalytic active sites of the electrode (Supplementary Material S2). Meanwhile, nanomaterials also exhibit excellent electroreduction and oxidation activity towards analytes, exhibiting better electrochemical performance. Multiple SPCE modified with different nanomaterials form a sensor array, which exhibits non-specific cross sensitivity to complex solutions and can express rich

* Corresponding authors.
E-mail addresses: jingjing_liu@neepu.edu.cn (J. Liu), qjliu@zju.edu.cn (Q. Liu).

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information contained in complex solutions. Among different kinds of electronic tongues, the voltammetric electronic tongue developed using the voltammetric principle is very popular. It possesses the advantages of rapid analysis, high signal-to-noise ratio and ease of use [11,12]. Furthermore, the voltammetric electronic tongue is considered a tool for obtaining comprehensive electrochemical information for the samples. Voltammetry excites the electroactive species in the sample under a controlled potential, while dynamically sampling the current, generating a large amount of information [13,14]. Especially for complex systems with multiple components, a single low-selectivity sensor in the voltammetric electronic tongue acquires the electrochemical information of some components in the mixed solution [15,16]. With the help of multiple sensors of different sensitivities, the complementary information provides a richer and more comprehensive description of the individual component [17,18].

While the voltammetric electronic tongue acquires rich data in complex solutions, the similar redox potentials of the components in complex solutions and the interference of the substrate will form an overlapping phenomenon of peaks. Such overlapping peaks containing nonlinear factors will cause the researcher difficulties in resolving the voltammetric data, making it difficult to obtain valid information for individual component. Chemometrics and machine learning provide methods for qualitative identification and quantitative analysis of individual components by obtaining effective information from voltammograms using multivariate calibration techniques [19–23]. Among them, convolutional neural network (CNN) is gradually applied in voltammetric data processing [24–26]. Although CNN has shown advantages in processing voltammetric data for local feature extraction and pattern integration of voltammogram obtained using a single electrode, the difficulties in exploiting the long-distance interactions between individual electrodes cannot be ignored [27,28]. It is a matter of concern how to achieve not only full use of the local patterns of each peak from a single electrode but also an effective correlation of the response patterns of each electrode to the same analyte to more efficiently extract the qualitative and quantitative information of analytes from voltammograms with severe signal overlap and nonlinear effects. In recent years, self-attention mechanisms have played an important role in natural language processing and image recognition tasks with the ability to establish global interaction [29–31]. It enables the global interaction of multiple vectors, adjusts the correlation of the vectors with the attention scores, and then captures the global dependencies [32]. This ability to encode from a global scale gives the possibility to correlate the response information of different electrodes to the same component. At the same time, the efficient fusion of the voltammetric response characteristics of each sensor in the voltammetric electronic tongue can further help to improve the recognition accuracy of the voltammetric pattern of multi-component solution.

Therefore, this work proposes a Fused Convolutional Transformer model (FCvT) that fuses convolution and self-attention to process voltammetric data. FCvT makes full use of the local features of the response of single sensor and the interconnected information between each sensor to improve the accuracy of the quantitative analysis of the individual component of complex mixtures by the electronic tongue. First, the voltammetric response of each sensor forms a data matrix in the form of independent vectors. The original matrix is convolutionally encoded by FCvT to form multiple voltammetric eigenvectors. The extraction of local peak patterns and the interaction of inter-sensor voltammetric features are then implemented in a Transformer block that combines depthwise separable convolution and self-attention. Finally, the high-order features obtained by the layered structure are converted into quantized values of components by the multilayer perceptron. FCvT is applied in the quantification of the component concentration of the mixed solution with 4 water quality parameters using the voltammetric electronic tongue. At the same time, the quantification error and model stability of the FCvT are compared with the traditional chemometric model and CNN model. The overall evaluation indicates that FCvT leads to improved accuracy for quantification.

2. Material and methods

2.1. Sample preparation

Based on the main parameters of real surface water, four substances were used to synthesize water samples for relevant experiments [33,34]. The composition of synthetic water shares some parameters with real surface water, and synthetic water samples are also known as mixed solutions.

The standard solution for preparing the mixed solution was provided by the Institute of Environmental Standard Materials of the Ministry of Environmental Protection (Beijing, China), which are $C_8H_5O_4K$ (9.53 mg/L), $NaNO_3$ (100 mg/L), KH_2PO_4 (100 mg/L) and NH_4Cl (100 mg/L). The standard solutions were diluted with deionized water so that each analyte was subjected to a full factorial experiment at three concentration levels (Table 1), resulting in a total of 81 ($3 \times 3 \times 3 \times 3$) mixed solution samples (supplementary material Table S1). The concentration of the analyte is determined comprehensively based on the Chinese surface water environmental quality standards and the indicators obtained from univariate calibration in previous studies [35].

2.2. Voltammetric electronic tongue system

The voltammetric tongue device used in the experiment consists of SPCE modified with nanomaterials, an electrochemical workstation, a multi-channel selector, and a computer (Fig. 1). Commercial SPCE were purchased from Xenslet studio in Ningbo, China, and the counter and working electrodes were carbon electrodes. The reference electrode was Ag/AgCl. Gold nanoparticles (AuNPs), silver nanodendritic structures (AgNDs) and reduced graphene oxide (rGO) were used to modify three SPCE to form a voltammetric sensing array, named AuNPs-SPCE, AgNDs-SPCE, and rGO-SPCE respectively (supplementary material S1). Electrode arrays were electrochemically detected using an electrochemical workstation (CHI760E, Shanghai CH instruments) and a multichannel selector drive (CHI684E, Shanghai CH instruments).

Nanomaterial modified SPCE are used for voltammetric electron tongue. These sensors can detect ions or covalent substances, electroactive or inactive compounds through various types of physical, chemical, and electrochemical interactions [36,37].

2.3. Sample processing

All samples of the mixed solution were measured in an optimized conditions [35]. The differential pulse voltammetry parameters for the three electrodes were similar, with an incremental potential of 4 mV, a pulse amplitude of 50 mV, and a pulse width of 20 ms. The only difference was the potential range, the scan range of AuNPs-SPCE and rGO-SPCE was between −0.8–0.8 V, and the scan range of AgNDs-SPCE was −0.8–0.3 V. The sensing array was replaced after every 30 measurements (Characterization of modified electrodes in S2 of supporting materials.).

Eighty-one mixed solutions were measured in a restricted random manner, and each mixed solution sample was subjected to 3 repeated measurements, resulting in a total of 243 (81×3) sets of measurement data, which constituted the total sample set [38,39]. To objectively

Table 1
Concentration levels of the four analytes in the mixed solution.

	$C_8H_5O_4K$ (mg/L)	$NaNO_3$ (mg/L)	KH_2PO_4 (mg/L)	NH_4Cl (mg/L)
Level 1	3.33	0.50	0.10	0.50
Level 2	5.00	1.00	0.20	1.00
Level 3	6.67	1.50	0.30	1.50

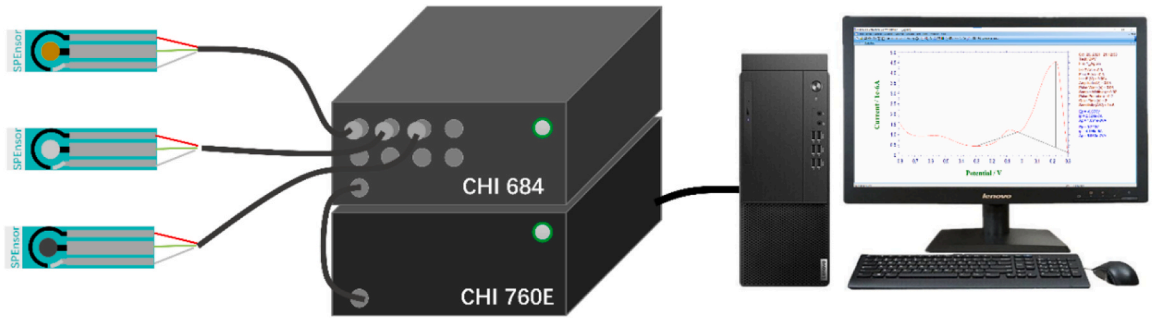


Fig. 1. Three kinds of nanomaterials modified screen-printed carbon electrode arrays and electrochemical workstations.

evaluate the performance of the model, the test subset was obtained from the random sampling of each analyte concentration level that follows the Gaussian distribution (Fig. 2). Finally, 16 mixed solution samples were classified into the testing subset (19 %, measurement data: $16 \times 3 = 48$), and the remaining 65 mixed solution samples constituted the training subset (81 %, measurement data: $65 \times 3 = 195$). Randomly select 20 % of the data in the training subset as the validation subset to adjust the hyperparameters of the model. Combinations of analytes from the separated test subset do not appear in the training subset, ensuring validity and accuracy when using the test subset to assess the model's generalization ability.

Data preprocessing and model design were done in the environment of Spyder IDE (Python 3.9) using the algorithm toolkit SciPy 1.4, machine learning library Scikit-learn 0.23, and machine learning platform TensorFlow 2.8. The voltammetric responses of the three electrodes were equalized by edge zero-filling, and then a voltammogram matrix (399×3) was formed according to the potential position. A Butterworth low pass filter filtered out high-frequency noise in each electrode response. The designed network built a model to quantify each analyte by preprocessing the completed training subset. It evaluated the model's generalization ability and real performance based on its performance in an independent test subset.

2.4. Data processing

Electrochemical measurements often contain various types of noise from the instruments, environment, etc., which cause spurious fluctuations in the signal [38,40]. The blue line in Fig. 3(a) shows the raw voltammetric data affected by noise coupling. Raw voltammograms were denoised using a Butterworth low-pass filter to highlight valid electrochemical signals. Among them, the low-pass cut-off frequency 0.3 *rad/s* of the response current signal was generated by AgNDs-SPCE, and the cut-off frequency of the response signal of AuNPs-SPCE and rGO-SPCE was 0.2 *rad/s*. The yellow lines in Fig. 3(a) show the filtered voltammetric data, and the significant effect of filtering can be seen from the locally enlarged image.

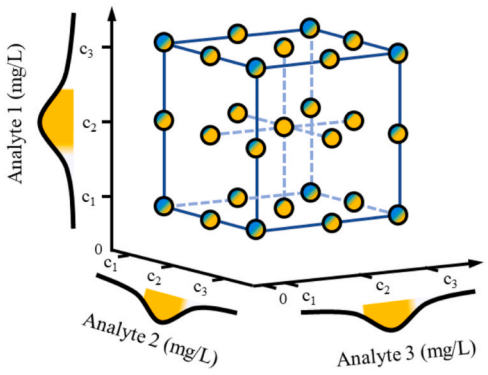


Fig. 2. Illustration of the combination of full factorial experimental designs for any three analytes and the distribution of training subsets and test subsets in the total sample set. Blue represents the training subset and orange represents the testing subset. The proportion of color in the circle represents the probability of the sample being classified into the corresponding subset in random sampling. Analyte 1, 2, 3 represent any three of the four analytes.

The combination of voltammetric response signals from electrodes needs to be considered from two main aspects: (1) Due to cross-sensitivity, the voltammogram of each electrode contains response information of different resolutions for multiple analytes, with high independence. (2) The relative position of the current peak and the peak potential in the voltammogram of the array sensor is the key information for the analyte species. Therefore, the response currents of the three electrodes are independent and are combined at the channel dimension in the form of electrode voltammetry vectors. At the same time, the current response of each channel maintains the relative positional relationship during the combination process. Due to the different effective potential windows of each electrode (AgNDs-SPCE: −0.75 to 0.15 V, AuNPs-SPCE: −0.7 to 0.7 V, rGO-SPCE: −0.8 to 0.8 V), the data dimensions are inconsistent. Therefore, edge 0 value padding is used to achieve equal length for each channel. Fig. 3(b) shows the combined results. Ultimately, the single measurement data forms a 399×3 (response value dimension × channel dimension) voltammogram matrix. 243 sets of measurement data constitute a complete data set of $243 \times 399 \times 3$.

2.5. Fused convolutional transformer design

A Transformer model that integrates convolution and self-attention mechanisms is proposed for processing voltammetric data obtained from a sensing array. This model adopts a convolutional encoding layer, two stacked fused convolutional Transformer blocks (FCT), a global maximum pooling layer, and a multi-layer perceptron. Fig. 4 shows the model structure of FCvT for voltammetric tongue data processing, and Table 2 lists the detailed configuration of each part of the model.

The model uses regular one-dimensional convolutions to form the encoding layer. It divides the current response signals of multiple electrodes into voltammetric sub-vectors through a large-span convolution operation. Voltammetric sub-vectors are feature vectors formed by the convolutional encoding layer after capturing the local information in the original data. Then, the encoded electronic tongue data enters the FCT block. The model exploits the properties of convolutional and self-attention mechanisms in a hierarchical structure by stacking of the FCT blocks. From a detailed point of view, the self-attention mechanism in the FCT block acts on the sub-vectors that have completed the feature mapping through the encoding layer. The contextual relationship of each sub-vector is obtained and the correlation at the global level is fully considered. Second, a convolution-based anti-bottleneck design utilizes a multilayer structure to sequentially capture the patterns ranging from simple low-level peak-shape features to higher-order voltammetric fingerprint patterns. Two FCT blocks are stacked in the design to reduce

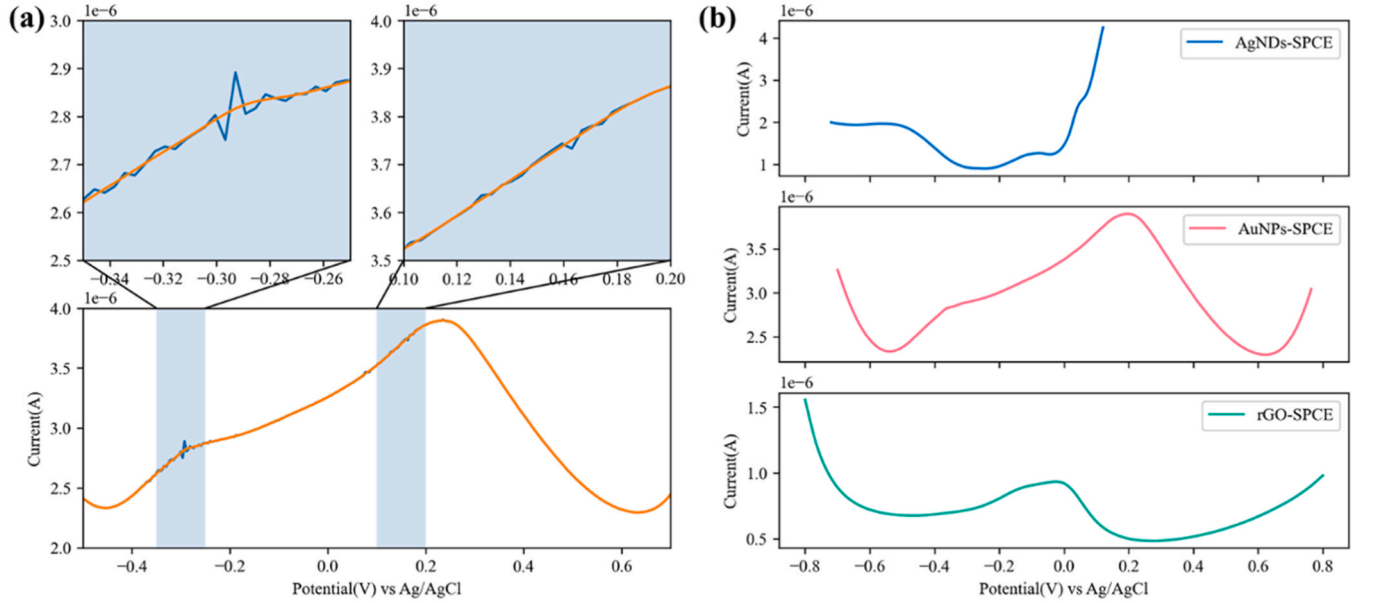


Fig. 3. Voltammetric data preprocessing (a) raw voltammetric data and filtered data, (b) composition of the three-electrode voltammetric data.

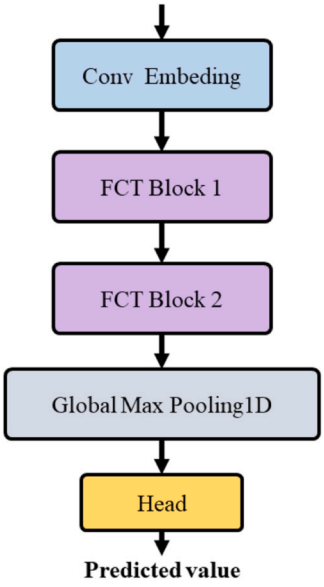


Fig. 4. Structure diagram of the quantitative regression model FCvT built using the FCT block. The voltammogram is input into the model after convolutional encoding, and the final predicted value is obtained through a two-stage FCT module and a head composed of multi-layer perceptron.

the parameters added to the model while taking advantage of the hierarchical structure (the total number of parameters in the model is only 12.94 K), reducing the obstacles of overfitting caused by the model in the case of limited data, and maintaining good generalization ability. After the data is abstracted by the FCT block, the maximum feature points are obtained by global maximum pooling to reduce redundant information. Finally, the data is transformed by the Head of the multi-layer perceptron using the Gelu activation function to obtain the predicted concentration of the analyte in the mixture.

2.6. Fused convolutional transformer block

The information acquired by the voltammetric tongue consists of the voltammetric responses of multiple electrodes with cross-sensitivity.

Table 2
Model parameter configuration for the mixed solution parameter regression task.

	Layer Name	Configure	Output Size	Activation Function
Input Embeding	Conv. Embed.	Filters = 20 Kernel size = 20 Strides = 20 Padding = 0	19×20	-
Block 1	Conv. Proj.	Kernel size = 3 Strides = 1	19×20	-
	Multi-Head Attention	Head num. = 2		-
	Invert. Bottle.	$D_1 = 20$ k = 2 $D_2 = 20$		Gelu
Block 2	Conv. Proj	Kernel size = 3 Strides = 1	19×20	-
	Multi-Head Attention	Head num. = 2		-
	Invert. Bottle.	$D_1 = 20$ k = 2 $D_2 = 20$		Gelu
-	Global Max Pooling1D	-	64	-
Head	Linear	64	64	Gelu
	Linear	4	4	
Params		12.94 K		

The input data is a 399 × 3 voltammogram. Conv. Embed.: Convolutional coding operation layer. Conv. Proj.: Convolution mapping operation. Invert. Bottle. The inverted bottleneck structure. Head num.: The number of heads for multi-head attention. D1: The output dimension of the multi-head attention mechanism. D2: The final output dimension of the inverted bottleneck structure. k: The multiplier of the dimension improvement in the inverted bottleneck structure.

The fused convolutional transformer block was designed to efficiently obtain valuable features related to analyte properties from the comprehensive voltammetric data (Fig. 5a). The self-attention

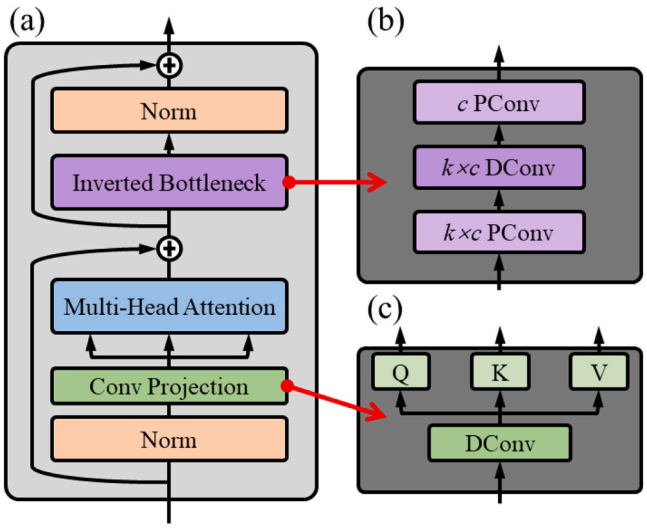


Fig. 5. Fused Convolutional Transformer block for the processing of voltammetric tongue data model.

mechanism transcends the distance limit and directly connects the individual voltammetric eigenvectors in the high-dimensional voltammogram. This not only makes full use of the current peak characteristics of multiple analytes at a single electrode, but also enables the interaction of peak characteristics for the same analyte in the global range of multiple electrode responses, rather than being limited to the acquisition of isolated peak characteristics.

The global interaction of self-attention on voltammetric signals helps enrich the expression of electrochemical patterns about analytes. However, self-attention mechanisms lack the ability to capture information in local region [41,42]. Therefore, in the design of FCT, the self-attention mechanism is supplemented by the efficient modeling ability of convolution under the local receptive field. The convolution captures the voltammetric peak shape characteristics under a limited receptive field in a sliding window manner, and the inductive bias of the space invariance and translation invariance makes the convolution maintain high efficiency in processing information based on positional relationships. In the FCT block, convolution operators are added in a loosely coupled manner. Convolutional projections of Q, K, and V are implemented using depthwise convolution (Dconv) (Fig. 5c). After the self-attention block, a depthwise separable convolution is added to form an anti-bottleneck structure (Fig. 5b) to extract the hidden features of the high-dimensional voltammetric data in the expanded dimension space, and then retain the effective features during the process of dimension reduction. It then models the global interaction information locally, reducing the ambiguity of the response current analysis within the potential region. The FCT block uses self-attention to obtain the global interaction relationship of the voltammetric response, and utilizes convolution for local feature extraction. It combines the characteristics of the two, avoiding their respective limitations, and forming an efficient aggregation mechanism for comprehensive analysis of voltammetric data formed by multiple electrodes.

2.7. Depthwise separable convolution

As a basic operator in various model designs, convolution has been proven to be efficient in various data processing tasks [43,44]. general convolution operations directly combine the relationship between the spatial dimension and the channel dimension of the feature map. Its operation is shown in formula (1).

$$Conv(W,x)_{(i,j)} = \sum_{m,n,c}^{M,N,C} W_{(m,n,c)} \cdot x_{(i+m,j+n,c)} \tag{1}$$

In formula (1), W is the weight matrix of the convolution kernel that can be trained, m, n and c correspond to the width, height and channel dimensions of the convolution kernel, respectively, and M, N and C also correspond to it. x represents the input feature map, i and j represent the position of the corresponding output feature map, respectively.

Different from regular convolution, depthwise separable convolution decouples the process of spatial information fusion and channel information fusion (Fig. 6). It separates the two fusions into two steps: Dconv only in the spatial dimension and pointwise convolution (PConv) only in the channel dimension [45]. For the electronic tongue data, the spatial information of the current at each potential contained in the voltammogram is captured by an independent convolution kernel, and the features are then fused across electrode dimensions with different sensitivities. The combination of these two links can avoid the generation of redundant features, extracting features more efficiently. Depthwise separable convolution can reduce the complexity of the model by reducing parameters, so that the model can maintain good generalization performance with the training of limited data sets.

DConv applies a single convolution kernel to each input channel, so the number of channels in the input and generated feature maps are the same. The operation of DConv can be expressed as:

$$DConv(W,x)_{(i,j)} = \sum_{m,n}^{M,N} W_{(m,n)} \cdot x_{(i+m,j+n)} \tag{2}$$

PConv can be regarded as a regular convolution with a convolution kernel size of one unit, which is used to fuse information in the channel direction of the feature map. PConv can flexibly adjust the number of channels to change the feature dimension. The process of PConv is as follows:

$$PConv(W,x)_{(i,j)} = \sum_{c}^{C} W_c \cdot x_{(i,j)} \tag{3}$$

The depthwise separable convolution greatly reduces the number of parameters by separating the two processes in the regular convolution. The ratio of the parameter used to those used by regular convolution operators is as follows:

$$\frac{m \times n \times c_i + c_i \times c_o}{m \times n \times c_i \times c_o} = \frac{1}{c_o} + \frac{1}{m \times n} \tag{4}$$

where, m, n are the size of the convolution kernel, c_i is the number of channels for the input feature map, and c_o is the number of channels for the output feature map.

The depthwise separable convolution can effectively obtain the local features of each electrode current response channel under the condition of minimized parameters, which reduces the possibility of overfitting in limited data sets.

2.8. Self-attention

Each electrode in the voltammetric sensing array separately provides cross-electrochemical information for each component in a multicomponent solution. The mechanism of self-attention is able to use the cross information in each electrode response to establish global dependencies of the voltammogram to better capture patterns of the complete voltammetric information. Fig. 7 shows the framework of self-attention.

Self-attention is a top-down attention mechanism, which can calculate the attention distribution of the input information according to the task situation, and generate the vector with the global interaction relationship [29]. The basic process of the self-attention mechanism is shown in Fig. 7. The input X is encoded into Query, Key and Value through W_Q, W_K and W_V weight matrices, respectively. Taking a sub-vector as an example, scaled dot-product calculation was performed on its Query and all Keys (including itself) to obtain the attention scores, and converts the attention score into a probability distribution under the action of the distribution function. Finally, the probabilistic score is multiplied by each Value and then summed to obtain the final output Z.

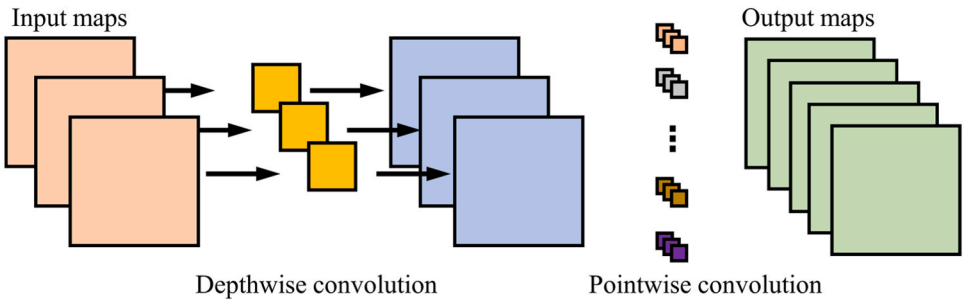


Fig. 6. The complete process of depthwise separable convolution.

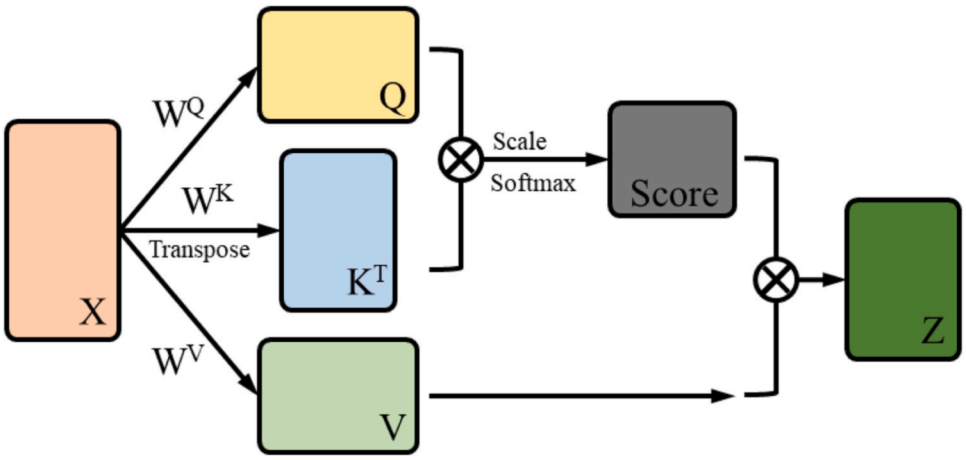


Fig. 7. Schematic illustration of the self-attention mechanism.

The main process of the attention mechanism is shown in Eq. 5, where d_k represents the dimensions of Q and K.

$$Attention\left(Q,K,V\right) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

On this basis, the multi-headed self-attention mechanism divides Query, Key and Value into multiple vectors of equal length, performs self-attention operation on each short vector separately, and finally connects the results. Each attention head can effectively pay attention to the information presented by the voltammetric array sensing data in different subspaces, and acquire specific meanings in the subspaces during the training process.

3. Results and discussion

3.1. Training settings and evaluation metrics

The training of the FCvT model was carried out under the Tensorflow framework version 2.8 in the Ubuntu 20.4 system. The hyperparameters of the model training process were set as follows: Batch size was 10, learning rate was fixed at 0.02, and number of iterations was 300. The loss function uses Huber (Formula 6), where $\delta = 1.0$. The optimizer chosen was Nadam [46].

$$L_\delta(y,f(x)) = \begin{cases} \frac{1}{2}(y-f(x))^2 & for|y-f(x)| \le \delta, \\ \delta|y-f(x)| - \frac{1}{2}\delta^2 & otherwise \end{cases} \tag{6}$$

Where y and $f(x)$ are measured and predicted values, respectively. δ is a parameter of Huber Loss.

The mean absolute percentage error (MAPE) (Formula 7) was used as an evaluation metric for the quantitative prediction results of the model, and the final model performance index was the average of 10 model results.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \widehat{Y}_i}{Y_i}\right| \times 100 \tag{7}$$

Where Y_i and $\widehat{Y}_i$ are measured and predicted values, respectively, and n is the number of samples.

3.2. Performance evaluation

To evaluate the effectiveness of FCvT, three regression methods commonly used in voltammetric tongue data processing were selected: PLS, SVR, and ANN. The parameters of each model were optimized respectively, and the appropriate number of hidden variables was determined by traversal method to obtain a PLS model with good generalization effect. The SVR model used the radial basis kernel function, and grid search to obtain the penalty parameter (C) and kernel parameter (γ) with the best prediction results. The ANN contained two hidden layers, and the activation function of each neuron was Gelu. At

Table 3
Prediction results of different methods for the concentration of four analytes in mixed solutions.

	Analyte 1 MAPE (%)	Analyte 2 MAPE (%)	Analyte 3 MAPE (%)	Analyte 4 MAPE (%)
PLS	4.57	10.85	44.31	13.25
SVR	3.61	9.39	52.08	12.07
ANN	2.25	6.98	49.60	13.82
CNN	3.96	4.74	52.87	10.70
FCvT	*1.50*	*3.30*	*22.26*	*8.06*

Analyte 1: $C_8H_5O_4K$; Analyte 2: $NaNO_3$; Analyte 3: KH_2PO_4; Analyte 4: NH_4Cl

the same time, a classic CNN was constructed as the basic model using the convolution operator for comparison. Table 3 lists the performance of each model for quantitative prediction of the four analytes in the mixed solution test set. It is not difficult to see that the FCvT model achieved the lowest MAPE values for all parameter predictions. Fig. 8 shows the results and stability of each model under multiple trainings in a more straight forward manner. Among them, PLS and SVR only show the model results under the best parameters.

From the MAPE of each model in Table 3, it is observed that as a commonly used PLS model for voltammetric tongue analysis, it is able to predict the concentrations of multiple analytes within an acceptable error range while maintaining ease of use. In contrast, the analytical methods SVR and ANN with nonlinear fitting ability performed better in the prediction results of some analytes. Among them, ANN, which is widely used in voltammogram analysis, was significantly better than SVR in the quantification results of analyte 1 and analyte 2. The CNN model utilizes the convolution operator to complement the ANN's lack of capturing ability for the local spatial information, and obtains lower quantification errors than ANN in the analysis of analytes 2 and 4. The proposed FCvT showed the best performance for predicting of all analytes in the mixture. FCvT aggregates the advantages of self-attention and convolution operators to overcome the limitation of overlapping peaks. Compared with CNN's quantization results for each parameter, its MAPE for prediction was reduced by 2.46 %, 1.17 %, 30.61 % and 2.64 %, respectively. The FCvT advantage is also reflected in the quantification of lower concentrations of analyte 3. The voltammetric peak of analyte 3 overlapped with the responses of other more prominent analytes, and the FCvT extracted valid features from complex voltammograms using a fusion mechanism of local and global information. FCvT showed better performance than other models, with reduced quantification error for analyte 3: 22.05%, 29.82%, 27.34% and 30.61% when compared to PLS, SVR, ANN and CNN, respectively. Finally, from the overall decline rate of quantization error ($\sum_{n=1}^{4}\frac{|error-error_n|}{4\times error_n}\times 100\%$, *error* is the MAPE value of FCvT, $error_n$ is the best value in other models), the average quantization error of FCvT for each component was reduced by 34.4% compared to the best results among other methods.

The error bars of ANN, CNN and FCvT in Fig. 8 respectively indicate the stability of the model. Compared to ANN and CNN, FCvT show good stability, while between CNN and FCvT, FCvT is better. ANN has a strong data-fitting ability, but the randomness of the initial weight increases the uncertainty of the learning process and makes the model less stable. The inductive bias inherent in the convolution operator becomes the key to improving the stability of the model. FCvT exhibits impressive performance in limited samples of voltammetric tongue data processing with a small number of parameters and a fusion processing mechanism.

4. Conclusion

In this paper, nanomaterial modified sensor arrays with cross sensitivity provide rich information on analytes in mixed solutions. A new model that fuses convolution and self-attention is proposed to overcome the difficulty of low-resolution voltammograms caused by peak overlap in voltammetric electronic tongue signal processing. The FCT block in the model fully integrates the interactive ability of self-attention in the global scope and the feature acquisition ability of the convolution operator in the local space. The body of the FCvT is then formed in a layered structure. FCvT is able to obtain efficient patterns related to the properties of the analytes in the complex voltammetric tongue data, thereby quantifying the concentration of each component

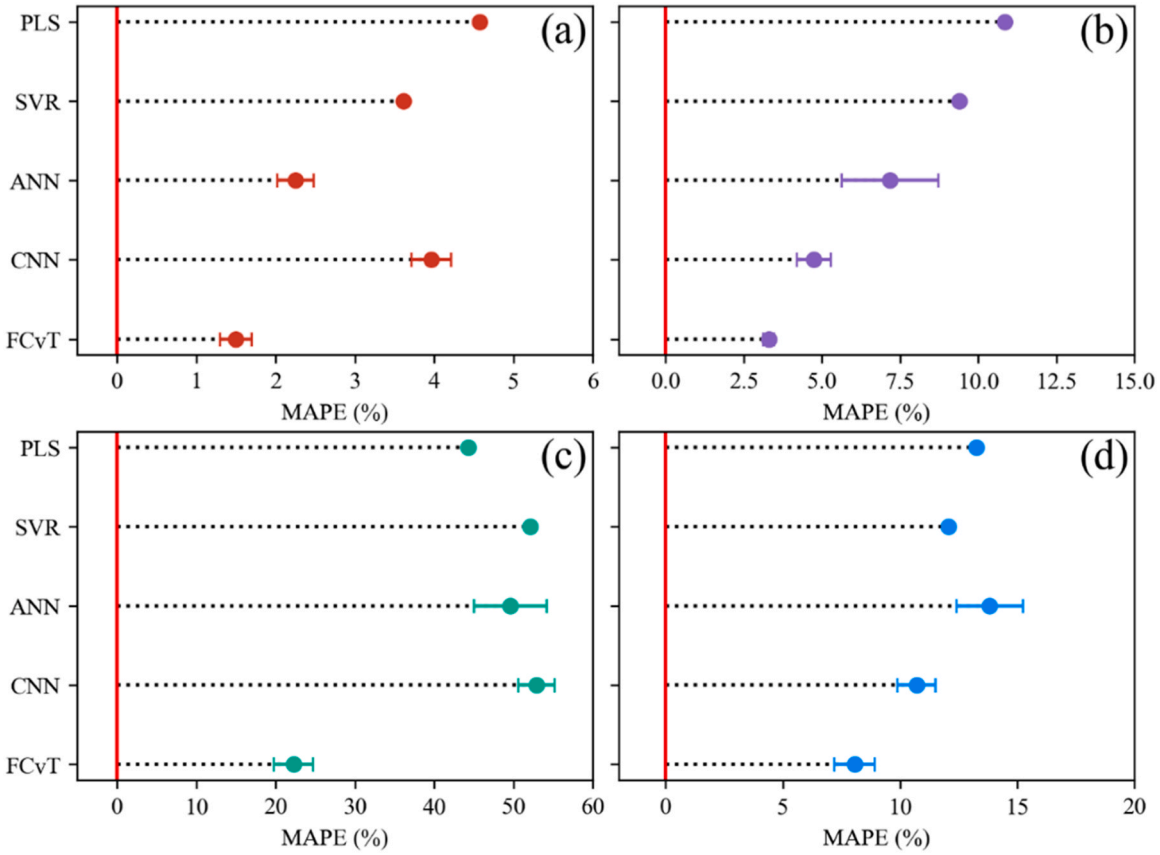


Fig. 8. Predicted results of different models for the concentrations of four analytes in mixed solutions. (a) to (d) are the mean absolute percentage errors of each model for the concentrations of analyte 1 ($C_8H_5O_4K$), analyte 2 ($NaNO_3$), analyte 3 (KH_2PO_4), and analyte 4 (NH_4Cl) in the test set, respectively. Error bars show standard deviation (n=10).

in the four analytes constituting mixed solutions. Compared to the commonly used models PLS, SVR and ANN in voltammetric electronic tongue processing, and the CNN model with gradual application in this field, FCvT shows excellent performance in the case of limited data set. In addition, FCvT takes into account the internal and contextual modes of voltammetric sub-vectors, and more clearly captures the information of low-concentration components in the voltammogram from the complex multi-electrode response signals, displaying great superiority. On the basis of electronic tongue voltammetry data processing, the FCvT efficiently integrates the characteristics of complementary information of multi-sensors, which can be well adapted to various tasks in multi-sensor systems, and has great application potential in quantifying certain parameters of real water quality.

CRediT authorship contribution statement

Qingjun Liu: Methodology, Conceptualization. **Sze Shin Low:** Resources, Investigation. **Haohao Ning:** Validation, Data curation. **Shixin Yu:** Writing – review & editing. **Dongfu Xu:** Validation, Supervision. **Jingjing Liu:** Methodology, Funding acquisition, Conceptualization. **Jiangyong Li:** Writing – original draft, Visualization, Methodology. **Yifei Xu:** Project administration, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2024.113462.

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